

Fast PPI STRING v12.5

Protein Pathway Simulator

Web Application User Manual

STRING API v12.5 | Weighted BFS Beam Search | Regulatory Network
Institute of AI for Emotion (AI4Emotion)

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1. Overview

Fast PPI STRING v12.5 is a web-based protein pathway analysis tool that reconstructs signaling pathways from phosphoproteomics time-series data using the STRING protein-protein interaction database (v12.5).

The application uses a BFS (Breadth-First Search) Beam Search algorithm with edge weights derived from STRING combined scores to find the most likely signaling pathways connecting phosphorylated proteins across time points.

1.1 Key Features

- STRING API v12.5 integration with three network modes (functional, physical, regulatory)
- Weighted BFS Beam Search with adaptive beam width for pathway reconstruction
- Regulatory network support: directionality (A→B) and signal type (activation/inhibition)
- Phosphorylation site tracking with GENE@SITE notation
- Novel protein sequence analysis via NCBI BLAST
- No software installation required — runs entirely in a web browser

1.2 How It Works

1. You upload a JSON file containing your phosphoproteomics data (start proteins, time-series targets, inhibitor drugs).
2. The simulator maps protein names to gene symbols via the STRING API.
3. For each time point, it performs BFS Beam Search to find connecting paths between the current pathway endpoints and the next set of target proteins.
4. Paths are ranked by cumulative edge cost (derived from STRING combined scores).
5. You download the resulting pathway files to your computer.

2. Getting Started

2.1 Accessing the Application

Open your web browser (Chrome, Firefox, Safari, or Edge) and navigate to the application URL provided by your administrator. No account creation or login is required.

NOTE: The application runs entirely in the cloud. All computation happens on the server; your computer only needs a web browser.

2.2 Interface Layout

The application interface is divided into three main areas:

Area	Location	Purpose
Settings Panel	Left sidebar	Configure network type, search depth, species, and advanced parameters
Input Area	Main area (top)	Upload JSON file or enter FASTA sequences
Control & Output	Main area (bottom)	Run/stop simulation, view logs, download results

2.3 Quick Start (5-Minute Guide)

6. Open the application URL in your browser.
7. In the left sidebar, confirm the default settings: Network Type = functional, Search Depth = 2, Species ID = 9606 (Human).
8. In the main area, click the "JSON File Upload" tab and upload your input JSON file.
9. Verify the file preview shows correct counts for Start Proteins, Time Points, and Inhibitor Drugs.
10. Click the green "Run Simulation" button and wait for completion.
11. When finished, scroll down to "Download Results" and click the download buttons to save each result file to your computer.

3. Preparing Input Files

3.1 JSON Input File Format

The primary input is a JSON file with three fields. The file must be saved with UTF-8 encoding.

Required Fields

Field	Type	Description
start_proteins	Array of strings	Receptor or initial signaling proteins (e.g., ["EGFR"])
time_series	Object { key: array }	Time-point data. Keys are labels (e.g., "t1", "t2"). Values are arrays of protein names.
inhibitor_drugs	Array of strings	Proteins to block from paths (e.g., drug targets). Can be empty [].

Example JSON File

```
{
  "start_proteins": ["EGFR"],
  "time_series": {
    "t1": ["SHC1@Y317", "GRB2", "SOS1"],
    "t2": ["HRAS", "RAF1", "-UBC"],
    "t3": ["MAP2K1", "MAPK1@T185_Y187"]
  },
  "inhibitor_drugs": ["PTPN11"]
}
```

Special Notation

Notation	Meaning
GENE@SITE	Phosphorylation site (e.g., AKT1@S473). Multiple sites separated by underscore: AKT1@S473_T308
-GENE	Negative/blocked protein. Prefix with a hyphen to exclude from paths at that time point (e.g., -UBC)
Time point keys	Can be any string. Sorted numerically if they contain numbers (t1, t2, ..., t10)

TIP: You can create this JSON file in any text editor (Notepad, VS Code, etc.). Save with .json extension and UTF-8 encoding.

3.2 FASTA Sequence Input (Optional)

For novel or uncharacterized proteins, you can provide amino-acid sequences in FASTA format. The application will attempt to map them to known STRING proteins using:

12. Name-based STRING mapping (fast, preferred)
13. NCBI BLAST homology search against SwissProt (slower, used as fallback)
14. Re-mapping BLAST hits back to STRING

FASTA Format Example

```
>MyNovelProtein
MRPSGTAGAALLALLAALCPASRALEEGQCRL
EEKKVNEEDAILAAVTLLNRLHSP...
>AnotherProtein
MAAPLSSPLMSKLSQLQEDLSQYKR...
```

NOTE: BLAST analysis can take 1-3 minutes per sequence. Use this feature only when name-based mapping is not possible.

4. Configuring the Simulation

All configuration options are located in the left sidebar panel.

4.1 Basic Settings

Setting	Default	Description
Network Type	functional	Type of STRING network to query (see Section 4.2)
Search Depth	2	Maximum number of intermediate proteins between two connected proteins in a path segment
Species ID	9606	NCBI taxonomy ID. Human = 9606, Mouse = 10090

TIP: A Search Depth of 2 means the algorithm looks for paths with up to 2 intermediate (hidden hub) proteins between each pair of consecutive time-point proteins. Depth 1 = direct neighbors only; Depth 3 = longer paths but much slower.

4.2 Network Types

STRING provides three types of protein interaction networks:

Network	Description	When to Use
functional	All evidence types: text-mining, experiments, databases, co-expression, genomic neighborhood, fusion, co-occurrence	General pathway analysis; broadest coverage
physical	Only experimentally verified physical protein-protein interactions	When you need high-confidence direct binding evidence
regulatory	Includes directionality (A→B) and regulatory type (activation, inhibition, catalysis)	When signal direction and activation/inhibition type matter

NOTE: Regulatory mode adds arrow notation in results: — (activation), — (inhibition), —→ (directed but type unknown).

4.3 Advanced Parameters

Click "Show Advanced Settings" in the sidebar to access these parameters. Default values work well for most analyses.

Parameter	Default	Description
Degree Cap H	50	Maximum number of neighbors

		considered per node during BFS. Lower = faster but may miss paths.
Beam Width K	150	Maximum number of candidate paths kept after each time point. Higher = more thorough but slower.
Max Paths per Target	300	Cap on paths found per source-target pair. Prevents explosion for highly connected proteins.
Min STRING Score	700	Minimum combined score (0-1000) for an interaction to be included. 700 = high confidence.
Hub Blacklist	(empty)	Comma-separated gene names to exclude (e.g., UBC, TP53). Useful for removing non-specific hub proteins.
Regulatory Weight Bonus	0.1	Weight reduction for direction-confirmed regulatory edges (regulatory mode only).
Verbose Logging	Off	Show detailed BFS progress in the execution log.
Collapse by Gene	On	Treat different isoforms of the same gene as identical.
No Immediate Backtracking	On	Prevent A→B→A patterns in paths.
Disallow Duplicate Genes	On	Prevent the same gene from appearing twice in a single path.

TIP: If the simulation is too slow, try reducing Degree Cap to 30 and Beam Width to 100. If too few paths are found, increase Min STRING Score threshold to 400 or increase Search Depth to 3.

5. Running a Simulation

5.1 Step-by-Step

15. Upload your JSON input file using the file uploader in the "JSON File Upload" tab.
16. After upload, verify the summary metrics: the number of Start Proteins, Time Points, and Inhibitor Drugs should match your expectations. Click "Preview JSON Contents" to inspect the data.
17. (Optional) If you have novel protein sequences, switch to the "Protein Sequence Input (FASTA)" tab and enter or upload your sequences.
18. Adjust settings in the left sidebar if needed (Network Type, Search Depth, etc.).
19. Click the green "Run Simulation" button.
20. A spinner will appear indicating the simulation is running. The Execution Log area will display progress messages.
21. When complete, the "Download Results" section will appear below with download buttons for each output file.

5.2 Execution Time Estimates

Simulation time depends on data complexity and settings:

Scenario	Typical Time
Small dataset (1 start, 3 time points, ~5 proteins each)	30 seconds – 2 minutes
Medium dataset (1 start, 5 time points, ~15 proteins each)	2 – 10 minutes
Large dataset (multiple starts, 8+ time points, 20+ proteins)	10 – 30+ minutes
With FASTA/BLAST sequence input	Add 1-3 minutes per sequence

WARNING: Very large datasets may exceed the server's time or memory limits. If this happens, try reducing the dataset size, lowering Search Depth, or reducing Beam Width.

5.3 Monitoring Progress

The Execution Log panel displays real-time progress including protein name resolution, BFS search progress per time point, the number of paths found and their costs, API call statistics and cache hit rates, and any warnings or errors encountered.

6. Understanding Results

6.1 Output Files

Three result files are generated after each successful simulation:

File 1: Complete Path Results (STRING_v12_results.txt)

Contains only fully connected pathways that span all time points from start to end. This is the primary result file. It includes a protein name-to-gene-symbol mapping table and, in regulatory mode, a regulatory interaction details table.

File 2: All Path Results (STRING_v12_all_results.txt)

Contains both complete paths and intermediate (partial) paths grouped by length. Intermediate paths are those that connected some time points but did not reach the final time point. Useful for understanding where pathway connections were lost.

File 3: Phosphorylation Site Paths (STRING_v12_all_Phospho_results.txt)

If your input data includes @SITE annotations, this file contains the same paths as above but with phosphorylation site information attached to each protein. For proteins with multiple sites, all site combinations are enumerated.

6.2 Reading Path Results

Functional/Physical Mode

```
Path #1: EGFR -> GRB2 -> SOS1 -> HRAS -> RAF1 -> MAP2K1 -> MAPK1
```

Each arrow (->) represents a protein-protein interaction from the STRING database.

Regulatory Mode

```
Path #1: EGFR — GRB2 — SOS1 —> HRAS — RAF1 — MAP2K1 — DUSP6
```

Arrow types indicate regulatory relationships:

Arrow	Meaning
—	Activation (or catalysis/expression)
—	Inhibition
—>	Directed (direction confirmed, but type unknown)
->	Undirected or direction not confirmed

Phosphorylation Site Paths

```
#1: EGFR -> GRB2 -> SOS1 -> HRAS -> RAF1 -> MAP2K1 -> MAPK1@T185_Y187
```

The @SITE notation shows which phosphorylation sites are associated with each protein at each time point.

6.3 Path Statistics

At the end of each result file, you will find summary statistics including the total number of complete and intermediate paths, network type used, elapsed time, and (in regulatory mode) counts of activation, inhibition, and directional edges.

7. Advanced Features

7.1 Using Hub Blacklists

Some highly connected proteins (hubs) such as UBC (ubiquitin C), TP53, or GAPDH can dominate pathway results without biological relevance to your specific signaling question. To exclude them:

22. Open the Advanced Settings in the sidebar.
23. In the "Hub Blacklist" field, enter gene names separated by commas (e.g., UBC,TP53,GAPDH).
24. These genes will be excluded from all BFS paths.

7.2 Custom Species

By default, the application analyzes human proteins (Species ID: 9606). To use a different organism:

Organism	Species ID
Human	9606
Mouse	10090
Rat	10116
Zebrafish	7955
Drosophila	7227
C. elegans	6239
S. cerevisiae	4932
Custom genome	STRG0AXXXXX (from STRING web upload)

For organisms with a custom genome uploaded to STRING, enter the assigned STRG0AXXXXX identifier in the Species ID field.

7.3 Combining JSON and FASTA Inputs

You can use both a JSON input file and FASTA sequences simultaneously. When both are provided, the start proteins from the JSON file and the FASTA sequences are merged (duplicates removed). The FASTA proteins are mapped to STRING using the name/BLAST strategy, and the JSON time-series and inhibitors are used as usual.

7.4 Tuning the Beam Search

The BFS Beam Search algorithm uses two key parameters that control the trade-off between completeness and speed:

- Beam Width (K): Controls how many candidate paths survive after each time point. A larger K finds more paths but uses more memory and time.

- Degree Cap (H): Limits the number of neighbors explored per protein. A lower H speeds up the search but may miss connections through less common interaction partners.

The algorithm also features adaptive beam width: when many distinct target proteins are detected at a time point, the beam automatically expands to maintain target coverage.

8. Troubleshooting

Problem	Cause	Solution
"Please upload a valid JSON file"	No file selected or file is corrupted	Re-upload your JSON file. Ensure it is valid JSON (check with jsonlint.com)
No paths found	Proteins not connected at the given score threshold, or incorrect protein names	Lower Min STRING Score to 400, increase Search Depth to 3, or verify protein names are correct gene symbols
Simulation is extremely slow	Large dataset with high Beam Width and Degree Cap	Reduce Beam Width to 100, Degree Cap to 30, or Search Depth to 1
Simulation timed out	Server time limit exceeded	Split your time-series into smaller chunks or reduce dataset complexity
Protein name not resolved	Protein name not recognized by STRING	Use standard HGNC gene symbols (e.g., EGFR, not ErbB1). Check at string-db.org
Rate limit error	Too many STRING API requests in a short time	The application automatically retries. Wait and try again if the problem persists
Blank or empty results file	All proteins excluded or blocked	Check your inhibitor_drugs list and hub blacklist; they may be blocking essential pathway proteins

9. Reference: JSON Input Format

9.1 Complete JSON Schema

```
{
  "start_proteins": ["PROTEIN_A", "PROTEIN_B"],

  "time_series": {
    "t1":  ["PROTEIN_C@Y100", "PROTEIN_D"],
    "t2":  ["PROTEIN_E", "-BLOCKED_PROTEIN"],
    "t5":  ["PROTEIN_F@S200_T300", "PROTEIN_G"],
    "t10": ["PROTEIN_H"]
  },

  "inhibitor_drugs": ["DRUG_TARGET_1", "DRUG_TARGET_2"]
}
```

9.2 Field Details

start_proteins

- Array of protein name strings
- These are the starting points (receptors) of the signaling pathway
- Usually 1 protein (e.g., ["EGFR"]), but multiple are allowed
- Each start protein initiates an independent pathway search

time_series

- Object where keys are time-point labels and values are arrays of protein names
- Time points are sorted numerically by any number found in the key
- Protein names can include @SITE notation for phosphorylation sites
- Prefix a protein name with "-" to block it at that time point
- Each time-point array represents the set of proteins activated/phosphorylated at that time

inhibitor_drugs

- Array of protein names to globally exclude from all paths
- These proteins are blocked at every time point (e.g., drug targets whose downstream signaling is inhibited)
- Can be an empty array [] if no inhibitors are used

10. Reference: Output File Descriptions

File	Contents
STRING_v12_results.txt	Name→Gene mapping table + complete paths only + regulatory details (if applicable)
STRING_v12_all_results.txt	Complete paths + intermediate paths grouped by length + path statistics summary
STRING_v12_all_Phospho_results.txt	All paths with phosphorylation site (@SITE) combinations expanded

10.1 Name-to-Gene Mapping Table

Located at the top of the results file. Shows how each protein name from your input was resolved to a STRING gene symbol. If a name could not be resolved, it appears as "None". Verify this table to ensure correct protein identification.

10.2 Complete vs. Intermediate Paths

- Complete paths: Successfully connected from a start protein through ALL time points to the final time point.
- Intermediate paths: Connected through some time points but stopped before reaching the end. These are useful for understanding where the pathway reconstruction encountered gaps.

10.3 Edge Weight and Cost

Each edge (protein-protein interaction) has a weight calculated as: $\text{weight} = -\ln(\text{combined_score} / 1000)$. Lower cost means higher confidence. Paths are ranked by total cumulative cost, so the top-ranked paths represent the highest-confidence signaling routes.

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